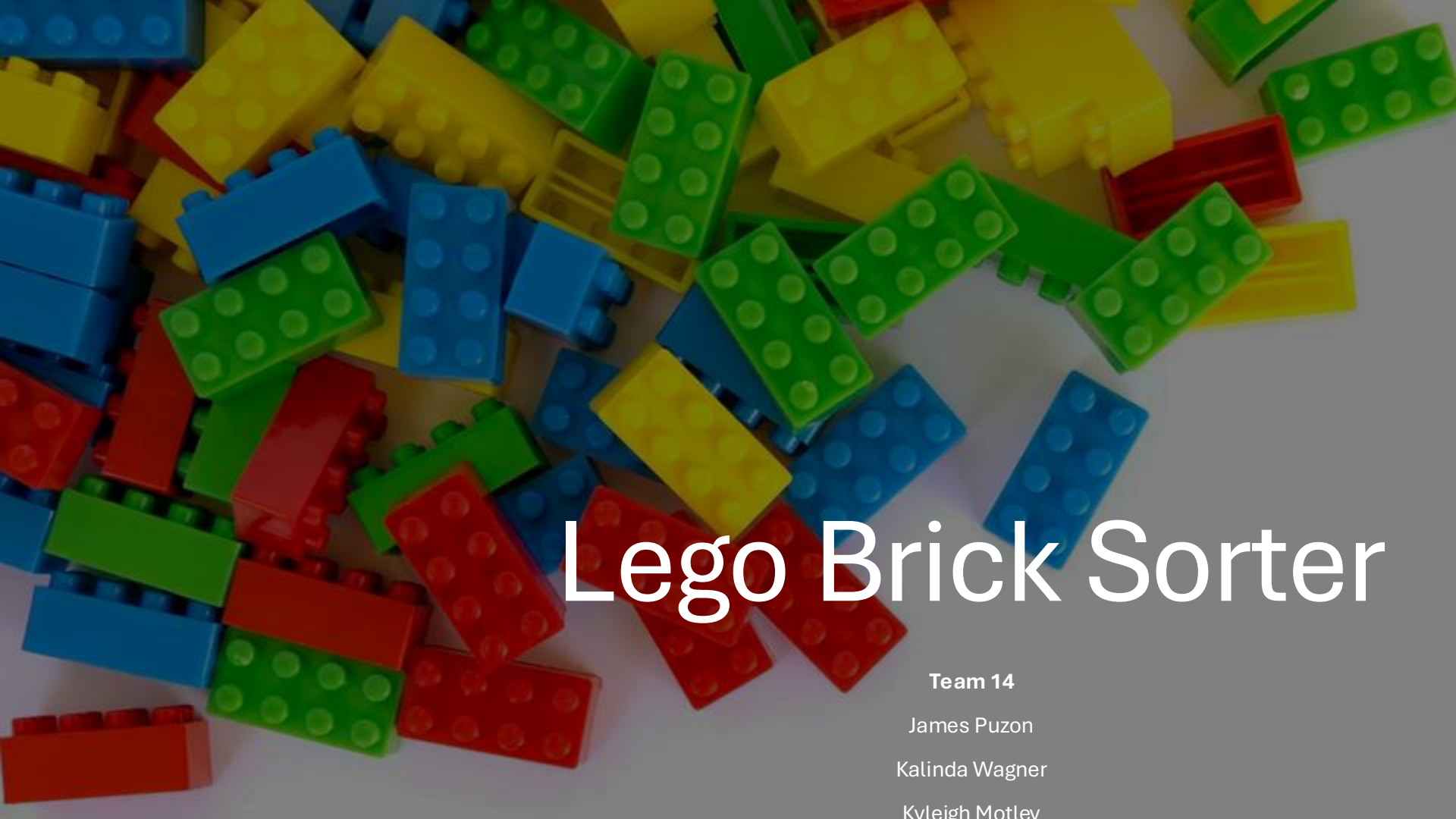




Lego Brick Sorter

Team 14

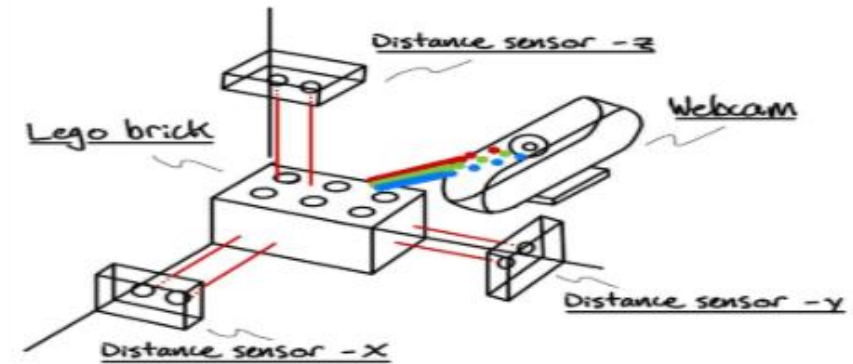
James Puzon

Kalinda Wagner

Kyleigh Motley

Project Overview

- Developed a system to identify LEGO blocks based on **geometric** and **visual** data
 - System has potential application of sorting LEGOs
- **Ultrasonic sensors + Arduino Uno:** measure block geometry (length, width, height)
- **Web camera + Python:** capture image and extract color information
- Determined the error associated with each measurement to assess capabilities of experiment



Preliminary Test Set-Up



Experimental Set-Up

Sensor Selections

- Selected **ultrasonic distance sensors**
 - Non-contact measurement of LEGO geometry
- All distance sensors interfaced via a single **Arduino Uno**
 - Power, data acquisition
- **Web camera** combined with **Python script**
 - Color identification

Sensor	Quantity	Manufacturer	Make/Model	Sensitivity
Ultrasonic Distance Sensor	2	Adafruit	4007	±0.5 [cm]
Ultrasonic Distance Sensor	1	Adafruit	HC-SR04	±0.5 [cm]
Web Camera	1	Logitech	C920s Pro HD Webcam	--

Full Equipment List

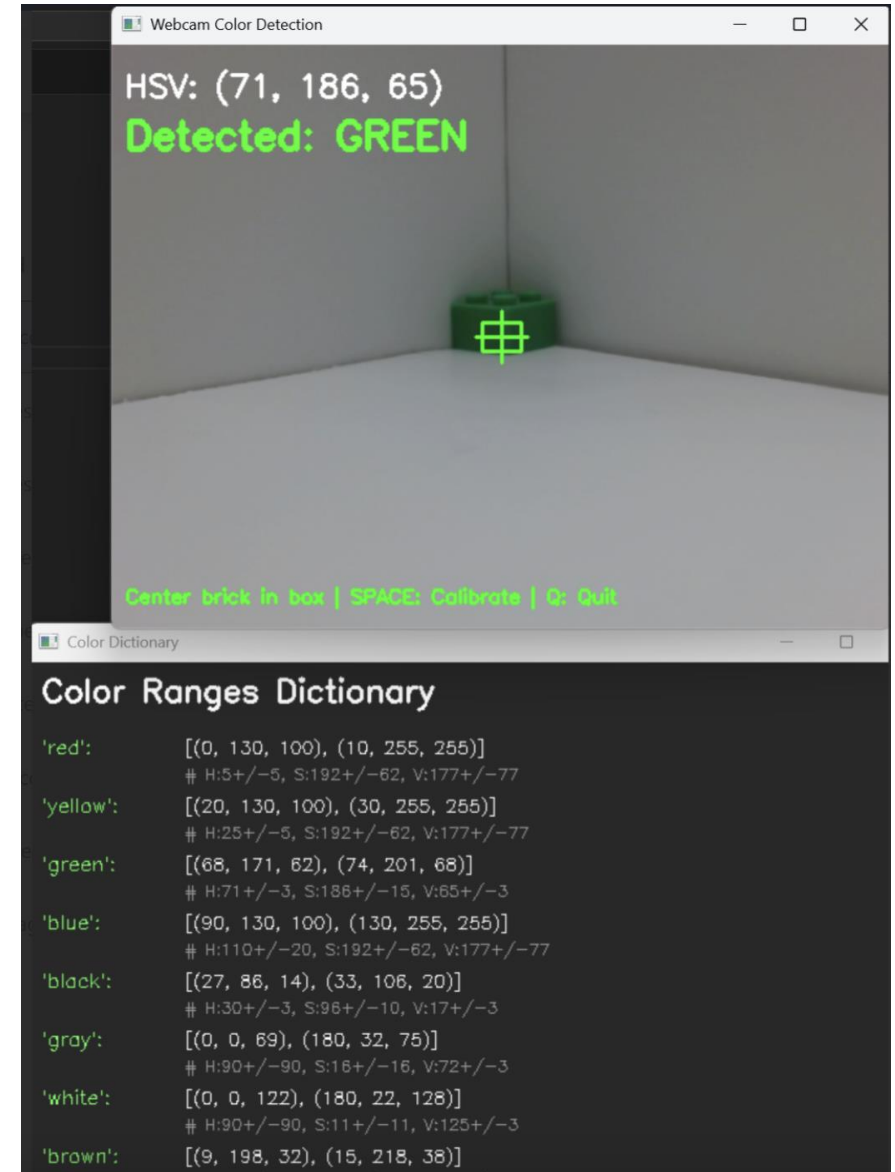
EQUIPMENT	QUANTITY	AVAILABILITY	COST
Distance Sensor (4007)	2	In lab	\$3.95 (per sensor)
Distance Sensor (HC-SR04)	1	In lab	\$3.95
Web Camera	1	From group member	\$54.99
Arduino	1	In lab	\$19.99
Wire	1 [ft]	In lab	\$5.00 (approx.)
Small Breadboard	3	In Lab	\$4.95
Poster board	1	From group member	\$5.00 (approx.)
Tape	8 [in]	In lab	\$2.00
Legos	20 (various sized pieces)	From group member	\$50.00 (approx.)
Ruler	1	In lab	\$4.50
TOTAL COST		\$162.19	

Color Recognition Calibration

- **Calibrates** dictionary of color ranges for accurate color classification within color identification script for conciseness
- **Process:**
 - Center LEGO block within defined region on screen
 - Script samples and outputs HSV pixel data
 - User inputs the corresponding color label
 - Script appends/replaces the HSV range + label to the color database
- Code available on GitHub
 - Link: <https://github.com/jamesPuzon/LegoBrickSorter>

Color Recognition

- A custom script in Python **identifies** the LEGO block's **color**
- **Process:**
 - LEGO piece placed within the camera's field of view
 - Script converts the image to **HSV** color values
 - Compares measured HSV values to saved color dictionary
 - Outputs the detected color
- Script includes list of colors and their corresponding HSV ranges
- Additional window displays color dictionary



Distance Sensor Calibration

- Sensors were calibrated using an Arduino sketch:
 - Sketch runs while measurement area is empty of LEGOs.
 - Sensors to collect data for 5 seconds
 - Data is averaged in each direction (x, y, and z)
 - The averaged values are put into the measurement script as the nominal values
 - Distance sensor to the wall

```
X: 31.80 cm, Y: 31.37 cm, Z: 28.59 cm
X: 31.79 cm, Y: 31.37 cm, Z: 28.59 cm
X: 31.79 cm, Y: 31.36 cm, Z: 28.59 cm
X: 31.91 cm, Y: 31.46 cm, Z: 28.71 cm
X: 31.79 cm, Y: 31.37 cm, Z: 28.59 cm
X: 31.80 cm, Y: 31.37 cm, Z: 28.59 cm
```

Data collected
over period of
seconds

```
--- Calibration Complete ---
Average Raw X Distance: 31.82 cm
Average Raw Y Distance: 30.87 cm
Average Raw Z Distance: 28.64 cm
```

Nominal distance
to wall in each
direction

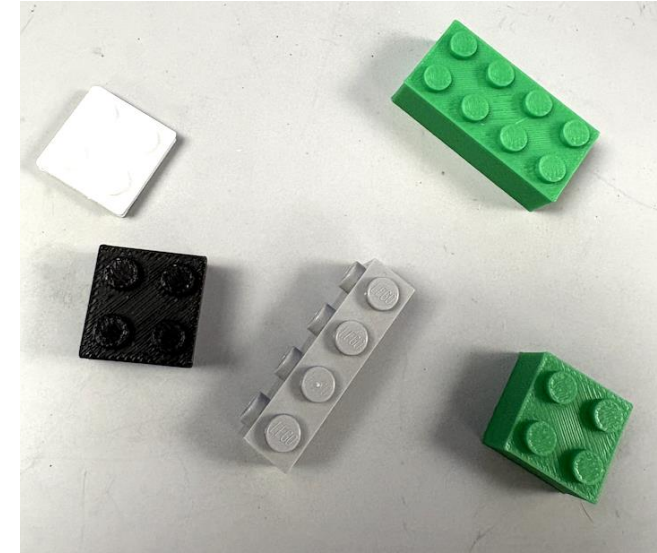
Distance Measurements

- Ultrasonic sensors determine distance by emitting and receiving sound pulses
- **Process:**
 - LEGO block is placed within the sensing range of all three sensors
 - Sensors fire a few milliseconds apart to avoid signal interference
 - For 10 seconds, each sensor records its distance to the LEGO
 - Distances are then subtracted from the nominal (i.e. no LEGO) distance
 - Final dimensions are obtained by averaging X, Y, and Z measurements
- Final dimensions measured were then recorded in Excel.

```
--- 10-SECOND AVERAGED RESULTS ---  
Avg Raw X: 31.07 cm  
Avg Raw Y: 29.05 cm  
Avg Raw Z: 28.29 cm  
X dimension: 0.94 cm  
Y dimension: 1.77 cm  
Z dimension: 0.82 cm  
-----  
Measurement complete.
```

Experimental Error

- A sample of five blocks was measured using the test and calibration procedures
- The percent and RMS errors were calculated for each LEGO as:
- In general, average absolute **error was high**
 - Indicates issues with sensor placement/setup and/or experiment design
 - **Specifically:** Global RMS + Average Height Error
- The accuracy of the color detection was 100%.



Average Length Error	23.24%
Average Width Error	16.19%
Average Height Error	113.65%
Global RMS Error	116.33%

Design of Experiments

How does lighting, background color, and camera quality affect how the camera reads the LEGO brick color?

Lighting: Dim (300 lumens), Bright (1500-2000 lumens)

Background color: White, Black

Camera Quality: Low (320x240), High (1920x1080)

Lighting	Background Color	Camera Quality
Dim	White	Low
Bright	White	Low
Dim	Black	Low
Bright	Black	Low
Dim	White	High
Bright	White	High
Dim	Black	High
Bright	Black	High

- **Simple, Low Cost Set Up**
 - Uses readily available components (ultrasonic sensors, Arduino, webcam)
 - Easy to assemble and modify
- **Non-Contact Measurements**
 - Ultrasonic distance sensors allow geometry measurement without touching the block
 - Helps reduce wear and alignment issues
- **Calibration**
 - Color recognition script can easily be updated to include new LEGO colors
 - Distance calibration includes using simple offsets
- **Easy to Modify Set Up**
 - Color detection and geometry measurement operate independently (use different sensors)
 - Components can be replaced without redesigning or impacting the rest of the system

Experimental Strengths

- **Geometric Constraints**

- System is optimized for rectangular or square blocks
- Distance sensors will still produce readings, but irregular shapes may cause incorrect classification without testing additional parameters

- **Color Detection Challenges**

- Web camera performance depends heavily on lighting conditions (poor lighting reduces color detection accuracy)
- Neutral colors (black, white, gray) difficult to distinguish reliably

- **Possible Sensor Interference**

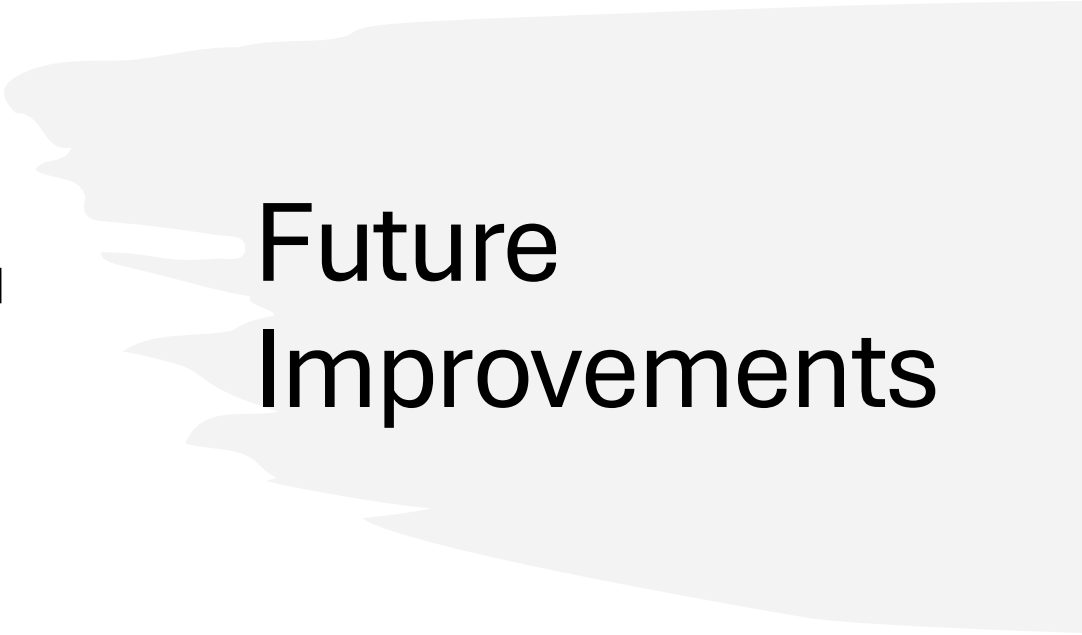
- Ultrasonic sensors may interfere with each other, especially those that are close together
- Even with small delays, echoes may bounce off walls causing inaccurate readings

- **Surface Material and LEGO Orientation Effects**

- Distance sensors don't work well with angled or irregular surfaces
- Even with rectangular blocks, studs on the top LEGO surfaces may lead to errors

Experimental Limitations

- **Use additional parameters for classification**
 - Weight, number of studs, surface texture
- **Improve lighting conditions**
 - Add LED lighting to reduce errors involving color recognition
- **Automation of testing**
 - Legos are currently placed by hand, but the process could be automated
 - For example, LEGOs may be placed with a ramp instead of by hand
- **Add more robust geometric sensing**
 - Infrared or laser distance sensors could contribute to higher accuracy for short range measurements
 - Ultrasonic sensors may interfere with each other in current set-up
- **Apply filtering to sensor measurements**
 - Apply filters to reduce noise and improve accuracy of measured geometry



Future Improvements